上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

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摘　要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以通过传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作（Vision-Language-Action, VLA）系统。系统设计运行于双臂 Franka 平台，并面向典型 PC 关键部件任务，包括内存条插入、CPU 放置以及电源线连接。

当前部署系统选择相应的 ACT 检查点，接收同步视觉观测和机器人状态，并在闭环执行过程中输出机器人及夹爪动作。现阶段证据限于干运行和有边界的验证活动；本文不声称已经实现语言条件推理、统一多任务控制、自主规划、故障检测、恢复或双臂协同操作。这些功能仍是面向预期 VLA 与双臂系统的长期扩展方向。

工程验收目标包括：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿误差低于 20 mm，故障检测率高于 90%，恢复成功率高于 70%。这些数值定义验证阈值，而非已经证明的性能结果。

关键词： 双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The deployed system selects the corresponding ACT checkpoint, receives synchronized visual observations and robot state, and outputs robot and gripper actions during closed-loop execution. Current evidence is limited to dry runs and bounded validation activities; the report does not claim demonstrated language-conditioned inference, unified multi-task control, autonomous planning, failure detection, recovery, or bimanual manipulation. These functions remain long-term extensions toward the intended Vision-Language-Action (VLA) and dual-arm system.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results.

Key words: dual-arm robot, vision-language-action model, industrial automation

Contents

Chapter 1 Introduction

1.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

Recent advances in Vision-Language-Action (VLA) models offer a promising route toward this goal. By jointly learning from visual observations, language instructions, robot proprioceptive states, and action data, VLA models can in principle generalize across tasks without requiring separate hand-engineered programs for every product variant. Landmark systems, including RT-2, OpenVLA, RDT-1B, and the π series, demonstrate the potential of transferring semantic knowledge from large vision-language models to physical robot control[1-4]. Earlier work on visual and language-conditioned manipulation also established complementary design patterns: Transporter Networks use spatially grounded pick-and-place representations, CLIPort combines semantic and spatial pathways, and PerAct couples language goals to structured visual observations[5-7]. In parallel, BridgeData V2, Open X-Embodiment, DROID, and LIBERO have made dataset scale, embodiment diversity, and reproducible task evaluation central considerations in robot-learning system design[8-11].

Despite this progress, adapting VLA-style systems to industrial assembly remains a system-level engineering problem. Many published demonstrations emphasize tabletop manipulation, single-arm control, or short-horizon tasks with relatively forgiving tolerances. In contrast, 3C assembly requires reliable perception of small components, accurate contact-rich manipulation, synchronized sensing, safe robot execution, and repeatable validation under bounded physical conditions. These requirements motivate a complete manipulation pipeline that links hardware selection, teleoperation, data collection, model training, deployment, and quantitative evaluation.

1.2 Design problem

This project aims to prototype a dual-arm industrial manipulation platform that couples camera observations and robot proprioceptive state to learned task actions for a scoped set of PC assembly operations. The target workcell consists of two Franka Research 3 (FR3) robot arms with complementary global and wrist/side camera views. Demonstrations are collected through a GELLO-based teleoperation interface and are recorded as synchronized visual observations, robot states, and robot or gripper actions. The intended task domain covers representative PC key-component operations, including RAM insertion, CPU placement, and power-cable connection.

At the present stage, task-specific Action Chunking with Transformers (ACT) checkpoints are selected by the operator rather than inferred from a language command[12]. The deployed system receives synchronized observations and robot state, then outputs robot and gripper actions for closed-loop execution. This implementation is therefore best understood as a staged imitation-learning baseline and a foundation for the intended VLA system, rather than as a fully demonstrated language-conditioned, unified multi-task controller. Language-conditioned inference, autonomous planning, failure detection, recovery, and robust bimanual coordination remain long-term extensions toward the final VLA and dual-arm manipulation concept.

1.3 Project objectives and scope

The project objective is to establish an end-to-end engineering pipeline for bounded industrial manipulation experiments. This pipeline includes hardware integration, synchronized multi-camera sensing, teleoperated demonstration collection, dataset preparation, ACT-based policy training, policy deployment, and repeatable physical validation. The expected Design Expo outcome is a measurable real-robot demonstration of the scoped assembly step, including task success, pose-estimation performance, and failure-handling evaluation.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results. Current evidence is limited to dry runs and bounded validation activities; therefore, this thesis does not claim unrestricted open-world generalization, autonomous factory deployment, guaranteed safe recovery from every failure mode, or fully demonstrated VLA-based multi-task inference.

1.4 Main contributions

The main contributions of this thesis are as follows:

1. A dual-FR3 manipulation platform is specified and integrated with GELLO teleoperation, synchronized camera sensing, robot-state recording, and task-compatible end effectors and fixtures.
2. A reproducible data and learning pipeline is developed to convert teleoperated demonstrations into datasets for ACT-based training and real-robot deployment.
3. A scoped industrial assembly validation framework is defined for representative PC component tasks, with quantitative targets for planning, execution, perception, pose accuracy, failure detection, and recovery.
4. A traceable engineering process links customer requirements and quality function deployment to concept generation, module selection, implementation decisions, and validation criteria.

1.5 Organization of this thesis

The remainder of this thesis is organized as follows. Chapter 3 translates customer requirements into measurable engineering specifications and presents the quality function deployment. Chapter 4 describes functional decomposition, concept generation, and selection of the major hardware, sensing, teleoperation, end-effector, and policy modules. Chapter 5 details the final system design, data and control architecture, model-training procedure, and implementation plan. Chapter 6 presents the physical validation methods and experimental results. Chapter 8 discusses the strengths and limitations of the system, summarizes the conclusions, and identifies directions for future work.

Chapter 2 Introduction

2.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

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Chapter 3 Design specification

3.1 Customer requirements

Through discussions with the project sponsor and analysis of the target application, eight customer requirements were identified and assigned importance weights reflecting the sponsor's priorities. The requirements capture both functional goals and non-functional concerns, including robustness, safety, generalizability, cost, and reliability.

Table 3–1 Customer requirements and importance weights

No.	Customer Requirement	Weight
CR1	Complete assembly task successfully	0.18
CR2	High assembly precision	0.16
CR3	Robust to environment and disturbances	0.15
CR4	Safe and collision-free operation	0.14
CR5	Fast task execution	0.12
CR6	Easy to use and generalize	0.10
CR7	Cost-effective system	0.08
CR8	System reliable and recovery capable	0.07

The customer requirements were translated into engineering specifications through the QFD process rather than by assigning targets independently. For each customer requirement i and engineering specification j, the team assigned a relationship value $r_{ij} \in \{0, 1, 3, 9\}$ (none, weak, moderate, or strong) and calculated the unnormalized engineering priority as

$$I_j = \sum_i w_i r_{ij}, \tag{3–1}$$

where w_i is the customer-importance weight in Table 3–1. The priorities were then normalized to identify the technical characteristics that most directly protect the high-weight needs. CR1 and CR5 primarily drive ES1 and ES2; CR2 drives ES3 and ES4; CR3, CR4, and CR8 drive ES5 and ES6; and CR6–CR7 constrain the selected implementation to a usable, feasible, and maintainable route. This translation produces an auditable chain from the sponsor's language to acceptance tests and explains why the later concept search is organized around platform, teleoperation, data collection, gripper, and policy modules.

3.2 Engineering specifications

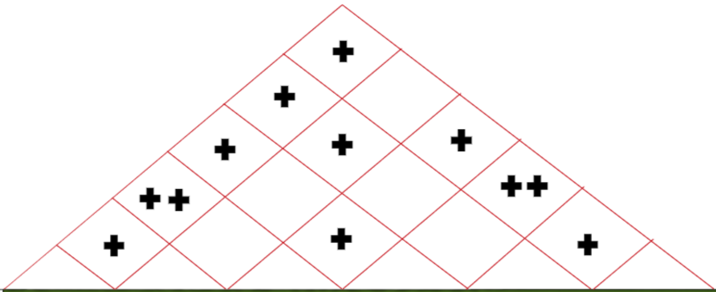
The customer requirements were converted into six measurable engineering specifications. The targets in Table 3–2 are the final acceptance thresholds for the present proof-of-concept, rather than performance claims. Each rate is calculated over independently initialized physical trials of the relevant task. A trial is counted once only; an unsuccessful recovery is not reclassified as an independent execution attempt. This convention prevents recovery behavior from artificially inflating the reported task-execution success rate.

Table 3–2 Final engineering specifications and acceptance thresholds

No.	Engineering Specification	Target	Operational Definition
ES1	Planning success rate	$> 90\%$	Fraction of trials in which the selected task policy and initialized scene form a valid executable plan.
ES2	Task execution success rate	$> 70\%$	Fraction of trials in which the prescribed assembly operation reaches its task-specific completion condition.
ES3	Component detection accuracy	$> 90\%$	Fraction of evaluated observations in which the required component is correctly detected for the active task.
ES4	Pose estimation error	$< 20\,\text{mm}$	Euclidean position error between the estimated and reference component pose.
ES5	Failure detection rate	$> 90\%$	Fraction of induced or naturally occurring execution failures that are identified before further unsafe action is issued.
ES6	Recovery rate	$> 70\%$	Fraction of detected failures for which the defined recovery procedure restores a valid task state.

3.3 Quality function deployment

Figure 3–1 records the resulting House of Quality. Its 9–3–1 relationship scale makes the engineering priorities interpretable rather than decorative: task completion, component localization, pose estimation, and failure handling carry influence across multiple high-weight customer requirements. The roof is used as a design-control device. For example, additional visual coverage can strengthen ES3 and ES4, but it increases calibration, synchronization, and data-management burden; similarly, recovery logic can improve ES5–ES6 while adding controller integration work. These QFD trade-offs define the subsequent concept space. The platform and gripper must enable precise contact motion for ES1–ES2 and ES4; the teleoperator and data pipeline must produce usable demonstrations and observable scenes; and the policy must convert those observations into safe executable actions while leaving explicit hooks for failure monitoring and recovery.



Customer Requirements	Weight (Importance)	Engineering Specifications (ES)					
		1. Task Execution Success Rate (%)	2. Pose Error (mm)	3. Failure Detection Rate (%)	4. Recovery Success Rate (%)	5. Component Detection Accuracy (%)	6. Planning Success Rate (%)
Complete assembly task successfully	0.18	9	3	3	3	3	9
High assembly precision	0.16	3	9	1	1	9	3
Robust to environment and disturbances	0.15	3	3	9	9	3	1
Safe operation	0.14	3	1	3	3	1	1
Fast task execution	0.12	3	1	1	1	1	3
Easy to use and generalize	0.10	1	3	3	3	3	9
Cost-effective system	0.08	1	1	1	1	1	1
System reliable and recovery capable	0.07	3	1	9	9	1	3
Absolute Importance (Sum)		372	314	360	360	314	394
Relative Importance (%)		17.6	14.9	17.0	17.0	14.9	18.6
Rank (1 = Highest)		2	5	3	3	5	1
Benchmark		85%	3.2 mm	85%	75%	92%	88%
Target		>70%	<20 mm	>90%	>70%	>90%	>90%

Illustration 3–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 4 Concept generation and selection

4.1 Concept generation

4.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

4.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables[13-14]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations[15-17]. These references define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

4.1.3 Morphological chart and generated module concepts

Table 4–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 4–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	**Dual Franka FR3**	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	**GELLO joint-master**	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	**Two wrist + one top camera + robot state/action**	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	**3D-printed soft gripper**
C5: Policy	**ACT in LeRobot**	π-style VLA policy	Diffusion policy

The morphological chart produces five leading module concepts, denoted C1–C5. C1 is a dual-FR3 platform; C2 is GELLO-based teleoperation; C3 is a synchronized three-camera, robot-state, and action pipeline; C4 is a 3D-printed soft gripper; and C5 is an ACT policy integrated through LeRobot. Together they form one integrated system concept, rather than five competing end-to-end robot systems. Their alternatives are retained because each exposes a real engineering trade-off: platform precision against integration cost, operator intuitiveness against mapping complexity, observation coverage against synchronization burden, compliance against geometric repeatability, and policy capability against data and deployment maturity. Detailed descriptions of the unselected alternatives are provided in the appendix.

4.2 Concept selection

4.2.1 Selection method and QFD-derived criteria

Selection was conducted in two stages. First, an absolute screen removed a candidate if it could not be made safe in the laboratory, could not interface with the selected robot and data path, or could not be obtained and integrated within the project schedule. Second, the surviving alternatives were scored on a five-point ordinal scale, where 1 denotes a weak fit and 5 denotes a strong fit. Ratings were assigned from the QFD relationship to ES1–ES6, laboratory availability and interface knowledge, the existing collection workflow, and the cited primary or official sources; no rating is presented as an unreported benchmark result. Consistent with the functional decomposition, the "top five concepts" are the five selected module concepts C1–C5. Their combination forms the single system concept carried into the final design. For candidate a in module m, the decision score is

$$S_{m,a} = \sum_{k=1}^{4} w_k r_{m,a,k}, \tag{4–1}$$

where $r_{m,a,k}$ is the rating for criterion k and w_k is its QFD-derived weight. Scores are used only to compare alternatives within the same module; they are recorded design judgments, not measured task-performance results.

Table 4–2 QFD-derived criteria for module selection

No.	Criterion	Weight	QFD Link
SC1	Contribution to task performance and precision	0.30	ES1, ES2, ES3, ES4
SC2	Data, control, and software integration compatibility	0.25	ES1, ES2, ES3
SC3	Safety, robustness, and recovery support	0.20	ES5, ES6; CR3–CR4
SC4	Availability, manufacturability, and schedule feasibility	0.25	CR6–CR7 and project plan

4.2.2 Scoring of the five selected module concepts

Table 4–3 provides the detailed selection record for the five leading module concepts. The same criterion definitions and weights are applied in every module so that the scoring logic remains traceable to the QFD; however, the totals are interpreted only within a module because a robot platform, a gripper, and a policy make different contributions to the integrated system.

Table 4–3 Module-level concept selection matrix (1 = weak fit; 5 = strong fit)

Module	Candidate	SC1	SC2	SC3	SC4	Weighted score
C1 Platform	Dual Franka FR3	5	5	4	3	**4.30**
	Dual Flexiv	4	3	4	2	3.30
	ALOHA-style bimanual	3	4	3	5	3.85
C2 Teleoperator	GELLO joint-master	4	5	4	5	**4.50**
	PICO 4 Ultra XR	3	3	3	4	3.25
	UMI hand-held	3	3	3	4	3.25
C3 Data collection	Two wrist + top camera + state/action	5	5	4	3	**4.35**
	One wrist + top camera + state/action	4	4	4	5	4.20
	Robot state/action only	2	3	3	5	3.25
C4 Gripper	3D-printed soft gripper	4	4	4	5	**4.20**
	Franka parallel gripper	3	5	4	3	3.70
	3D-printed hard gripper	4	5	3	4	4.05
C5 Policy	ACT in LeRobot	4	5	4	5	**4.50**
	Diffusion Policy	4	3	4	4	3.75
	π-style VLA policy (future route)	5	2	3	1	2.75

4.2.3 Comparative evaluation and selected concept

C1: Dual Franka FR3 platform. The dual Franka FR3 platform was selected as the mechanical foundation of the final system because it provides repeatable 7-DoF control, a direct connection to the available laboratory control ecosystem, and a credible path toward coordinated bimanual manipulation. Its principal disadvantage is the substantial integration effort required for workspace coordination, collision avoidance, commissioning, and bilateral control. Nevertheless, these costs are justified by the project objective: lower-cost alternatives may reduce initial hardware burden, but they would introduce a different control ecosystem or provide a weaker fit to the required assembly precision.

C2: GELLO teleoperation. GELLO was selected as the planned demonstration-collection interface because it offers a direct and intuitive connection between operator motion and the robot-state/action records required by the learning pipeline[18]. Despite the need for reliable leader–follower mapping and calibration, GELLO remains preferable to immersive or handheld alternatives for the present system because it provides the most direct route from teleoperated demonstrations to the joint-space data representation used by the FR3 and LeRobot workflow.

C3: Three-camera data pipeline. The selected sensing configuration uses one Intel RealSense D435 overhead camera and two Intel RealSense D405 wrist-mounted cameras, together with synchronized robot state, gripper state, and action records. The D435 provides a stable global view of the shared workspace, while the two D405 cameras preserve local visual information near the end-effectors and contact regions. This complementary arrangement reduces visual blind spots and directly supports the component-detection and pose-estimation requirements in ES3–ES4, at the cost of increased synchronization and data-management effort.

C4: 3D-printed soft gripper. The selected end-effector is a 3D-printed soft gripper derived from the open-source gripper design in the referenced UMI material; the project adopts only the gripper-finger component rather than reproducing the complete UMI hardware system[19]. Compliant contact and rapid geometry iteration are valuable for handling assembly components with small geometric variation, although deformation may reduce pose repeatability compared with a rigid locating gripper.

C5: ACT policy in LeRobot. ACT was selected as the current policy because it is integrated with the LeRobot workflow and supports task-specific training and checkpoint deployment from synchronized visual observations and robot-state inputs[12,20]. Action chunking provides a practical temporal-control mechanism for the present imitation-learning pipeline. Deployment remains task-specific and operator-selected: the policy does not yet infer tasks from natural-language input or demonstrate unified multi-task control. This limitation is accepted because ACT provides a deployable and traceable baseline for the current project stage, whereas Diffusion Policy remains an earlier baseline and π-style VLA models remain a future extension.

The selected architecture is the combination of the highest-scoring surviving candidate in each module. C1 and C4 establish the physical capability for controlled contact; C2 and C3 make demonstrations and visual evidence available to the learner; and C5 provides the deployable policy path. This combined decision covers the full ES1–ES6 chain while retaining a feasible implementation route for subsequent final design, manufacturing, and validation.

Chapter 5 Final design and implementation plan

5.1 Final design

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The final design follows the selected concept combination identified in Chapter 4. The system is built around a dual-Franka Research 3 workcell, a GELLO-based teleoperation interface, synchronized multi-camera sensing, task-compatible end effectors and fixtures, and an ACT policy deployed through the LeRobot workflow[12,18,20]. This architecture connects the mechanical platform, data path, policy model, and validation procedure into one end-to-end manipulation system.

The dual-FR3 platform provides the mechanical foundation for the completed bimanual manipulation task. The two arms operate in a shared, bounded workspace so that coordinated manipulation can be executed under controlled laboratory conditions. Low-level robot limits, controller protections, and workspace constraints remain active during execution. This arrangement satisfies the project need for bimanual manipulation while keeping the prototype within the defined proof-of-concept scope.

The demonstration and learning pipeline follows the data flow described in the DR3 report. Human demonstrations are collected through the GELLO leader device and converted into synchronized records containing camera observations, robot proprioceptive state, gripper state, and commanded actions. The top camera preserves global workspace context, while wrist or side camera views provide local visual information near contact regions. Episodes are screened for synchronization and task completion before being converted to the training format.

The policy module uses ACT as the current deployable controller. ACT predicts action chunks from visual observations and robot state, which provides a practical temporal-control mechanism for real-robot execution[12]. The implemented system therefore establishes a working bimanual imitation-learning baseline while preserving a route toward future language-conditioned policy selection and multi-task VLA deployment.

5.2 Design details

5.3 Implementation plan

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Implementation was organized around the same module boundaries used in concept generation. The hardware stage established the dual-FR3 workcell, mounted the task-compatible end effectors, and verified the available sensing layout. The teleoperation stage connected the GELLO interface to the robot-side data pipeline so that operator demonstrations could be recorded with consistent state and action channels. The data stage synchronized camera observations with robot state, gripper state, and actions, then converted accepted episodes into the learning dataset format.

The learning stage trained ACT checkpoints from the collected demonstrations and integrated the selected checkpoint into the real-robot execution loop. The deployment stage connected policy inference to the robot controller under bounded workspace and safety assumptions. The final integration stage completed the bimanual manipulation task at the system level, confirming that the hardware, sensing, teleoperation, dataset conversion, policy inference, and robot execution components can operate together in one closed-loop workflow.

Validation was planned as a staged process rather than a single final trial. Early checks focused on individual hardware communication, camera availability, state logging, and replay consistency. Later checks evaluated complete policy execution on the physical bimanual setup. This staged implementation reduces integration risk because failures can be traced to a specific module: robot control, teleoperation mapping, sensing, dataset preparation, model inference, or execution safety.

Chapter 6 Validation results

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6.1 Validation overview

The validation procedure follows the DR3 engineering specifications in Chapter 3. The purpose of validation is to check whether the integrated system satisfies the intended proof-of-concept behavior under bounded laboratory conditions. The evaluated workflow includes scene setup, synchronized sensing, state acquisition, policy inference, robot and gripper command execution, and completion assessment.

The bimanual manipulation task has been completed on the integrated dual-arm setup. This result demonstrates that the selected hardware platform, teleoperation-derived data path, ACT-based policy module, and robot-side execution loop can be combined into a functioning bimanual manipulation system. The result should be interpreted within the defined project scope: the workspace, initial conditions, fixtures, and execution assumptions are controlled, and the system is not claimed to provide unrestricted open-world autonomy.

6.2 Key data

6.3 Validation summary

The completed validation confirms the main system-level objective of the project: a dual-arm FR3 manipulation platform can execute the scoped bimanual manipulation workflow using the selected data and policy architecture. The most important validation outcome is the integration of the complete chain from demonstration collection and dataset preparation to learned policy execution on the real robot. Detailed task descriptions, trial counts, and quantitative measurements are reserved for the blank key-data section.

Chapter 7 Discussion and conclusions

7.1 Discussion

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7.2 Conclusions

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Chapter 8 Discussion and conclusions

8.1 Discussion

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This project developed a bimanual industrial manipulation system for representative 3C assembly operations. The final system integrates a dual-Franka Research 3 platform, GELLO-based demonstration collection, synchronized visual and proprioceptive data recording, task-compatible end effectors and fixtures, and an ACT-based policy deployment workflow. The design process remains traceable from customer requirements to engineering specifications, concept generation, concept selection, implementation, and validation.

A key outcome is that the bimanual manipulation task has been completed on the integrated real-robot setup. This confirms that the selected mechanical, sensing, data, and policy modules can operate together as a system rather than only as isolated components. The completed bimanual result also supports the original motivation of moving beyond single-arm demonstrations toward coordinated manipulation in a shared workcell.

The main limitation is that the present implementation remains a bounded laboratory proof of concept. The task setup, workspace, initial conditions, and fixtures are controlled. The ACT controller provides a deployable imitation-learning baseline, while language-conditioned checkpoint selection, unified multi-task VLA inference, autonomous planning, and generalized recovery remain outside the demonstrated scope.

8.2 Conclusions

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The thesis presents an end-to-end engineering process for VLA-oriented bimanual industrial manipulation. Starting from customer requirements, the project defined measurable engineering specifications, generated modular solution concepts, selected a dual-FR3 and GELLO-based architecture, and implemented a data-driven manipulation pipeline using synchronized sensing and ACT policy deployment.

The completed system demonstrates that a dual-arm robot workcell can be integrated with teleoperation, dataset generation, learning-based control, and real-robot execution for the scoped bimanual manipulation task. The contribution is therefore both a working prototype and a traceable design methodology: the final design choices can be related back to the QFD requirements, while the validation structure provides a basis for reporting task success, perception quality, pose accuracy, failure handling, and recovery once the detailed data section is completed.

Overall, the project establishes a practical foundation for flexible 3C assembly research. It shows that bimanual manipulation can be achieved within a controlled laboratory workcell using the selected hardware and learning pipeline, while leaving task-specific details and quantitative key data to be documented separately.

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Engineering changes notice Appendix 1

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Bill of materials Appendix 2

Table 2–1 Bill of materials

Quantity	Part Description	Purchased From	Part Number	Price (each)	Notes
...	...	...	...	...	...

Research Projects and Publications during Undergraduate Period

[1] ……

Acknowledgements

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NOx and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions. ……